

ASSIGNMENT 2-MAPREDUCE

Write a MapReduce functions for the following applications

1. Calculate the average marks obtained by each student.

Input Format

101,Math,80
101,Science,90
101,English,70
102,Math,60
102,Science,75
102,English,65

Output Format

101 80
102 66.67

2. Calculate the total salary paid to each department.

Input Format

101,HR,30000
102,IT,50000
103,HR,35000
104,IT,45000
105,Sales,25000

Output Format

HR 65000
IT 95000
Sales 25000

3. Calculate the attendance percentage for each student.

Input Format

101,Present
101,Present
101,Absent
102,Present
102,Absent

102,Present

Output Format

101 66.67%

102 66.67%

4. Count Even and Odd Numbers

Objective

Count the number of even and odd integers.

Input Format

10

15

18

25

30

45

Output Format

Even 3

Odd 3

5. Find the average rating for each movie.

Input Format

MovieA,5

MovieA,4

MovieB,3

MovieA,5

MovieB,4

Output Format

MovieA 4.67

MovieB 3.50

6. Identify the student with the highest marks in each department.

Input Format

CSE,101,85

CSE,102,90

ECE,103,88

ECE,104,92

CSE,105,80

Output Format

CSE 102 90

ECE 104 92

7. Identify the longest word in the input file.

Input Format

Artificial Intelligence Hadoop Spark Analytics

Output Format

Intelligence

8. Count the number of log messages for each log level.

Input Format

INFO System Started

ERROR File Missing

INFO User Login

WARNING Low Memory

ERROR Database Failed

Output Format

ERROR 2

INFO 2

WARNING 1

9. Get the maximum average temperature for every city.

Input format: City Name, Year, month1_Temp, month2_Temp, month3_Temp,
Avg_Temp_of_Year

1	chennai,2010,30,31,32,33
2	bangalore,2012,20,23,23,21
3	chennai,2016,37,38,37,35
4	coimbatore,2016,33,34,33,31
5	bangalore,2015,25,26,26,25
6	cochin,2016,29,30,28,27
7	chennai,2015,35,33,36,37

10. Sort the students name based on GPA

Input format: Student name, GPA

1	Dharun,8.6
2	Prithvi,9.2
3	Suriya,7.8
4	Arjun,8.5
5	Chandran,5.9
6	Indhran,9.0
7	Vishwanth,7.6

11. Reverse the contents of every input line.

Input

Hello Hadoop

Big Data Analytics

Map Reduce Programming

Output

poodaH olleH

scitylanA ataD giB

gnimmargorP ecudeR paM

12. **Convert to Uppercase**

Input

Hello Hadoop

Big Data

Output

HELLO HADOOP

BIG DATA

13. **Count Digits**

Input

ABC123

456DEF

789

Output

ABC123 3

456DEF 3

789 3

14. Remove Duplicate Words

Input

Apple

Apple

Orange

Apple

Banana

Orange

Output

Apple

Banana

Orange

15. Find Second Highest Salary

Input

Ram,40000

Ravi,55000

John,70000

Anu,60000

Output

Anu 60000